

MAIB AI 208 · AI IN MARKETING · PROJECT REPORT

RetentionLens

Ninety-day lapse risk and customer value for every wholesale account, the evidence behind each call, and outreach a person approves before anything leaves.

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Live demo: krish2105.github.io/Retention-Lens-AI-in-Marketing-

Code: github.com/krish2105/Retention-Lens-AI-in-Marketing-

What was built and what it shows

RetentionLens scores 5,852 trade accounts from a real UK giftware wholesaler (UCI Online Retail II, 1,067,371 invoice lines, CC BY 4.0) for the chance they place no order in the next 90 days and for their expected spend. It explains each call, groups accounts into segments, adds web and network signals, and drafts a check-in email with a local language model that a person must approve.

2,456	£324k	0.820	0.821
accounts at high lapse risk (≥ 75%)	usual 90-day spend at stake in them	PR-AUC of BG/NBD on the frozen test	AUC of the new-customer lookalike model

Main findings

- **Simple probabilistic models win.** Across four rolling cutoffs, LightGBM beat BG/NBD on 0 and logistic RFM on 1. A neural model is not justified by this evidence.
- **Wholesale buying leads consumer interest by about a month.** For Christmas lines the correlation with Wikipedia interest rises from 0.34 (same month) to 0.90 (one month later). Twelve monthly points: exploratory.
- **New customers can be spotted from their first order** (test AUC 0.821 on later customers).
- **A small local LLM needs a person in the loop.** 15/16 drafts kept to the model's numbers; its own second-pass review passed only 1/16.

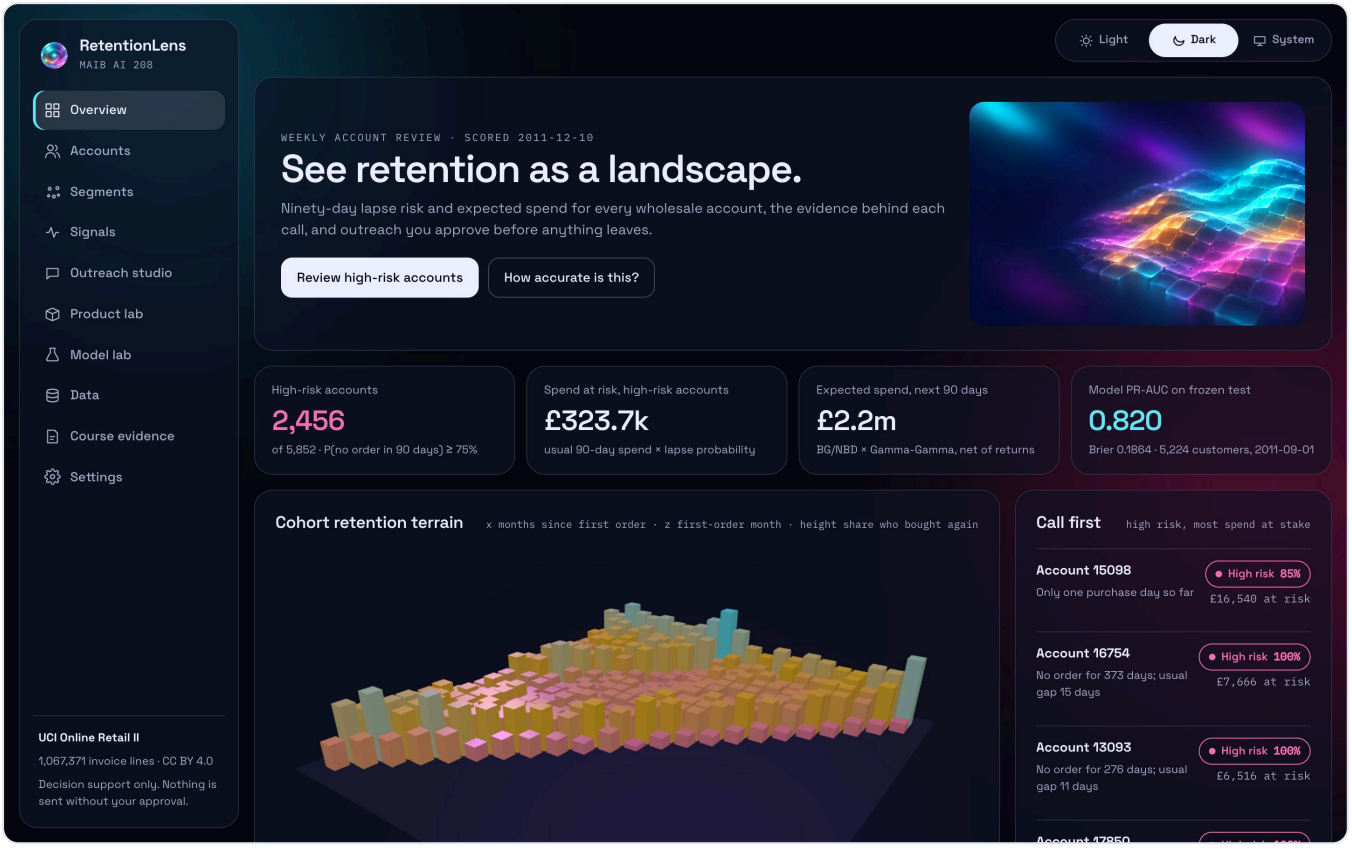


Figure 1. Overview screen (dark theme): key figures and the cohort retention terrain.

2 · PROBLEM AND USER

Who it is for and why it matters

The target user is the owner or account manager of a small wholesaler with hundreds of trade accounts and no data team. Between 65% and 67% of known accounts placed no order in the following 90 days at the development cutoffs, and lapses are usually noticed only after revenue has gone.

Desk research found a gap between self-serve direct-to-consumer tools (which need e-commerce events and hundreds of buyers) and distributor platforms (quote-only, ERP-integrated). No tool combined CSV-first input, small-sample uncertainty, approval-gated AI drafts and published pricing. Buyer, workflow and price are **hypotheses**: no interviews or pilot have been run yet.

3 · DATA

One licensed source, every problem counted

MEASURE	VALUE
Invoice lines	1,067,371
Date range	2009-12-01 to 2011-12-09
Invoices · customers · countries	53,628 · 5,942 · 43
No customer ID	243,007 (22.8%)
Cancellations · negative quantities	19,494 · 22,950
Duplicates (across sheets / within a sheet)	22,202 / 12,133
Rows usable for customer models	794,163 (74.4%), 5,875 customers

The two workbook sheets overlap on 1–9 December 2010, so the published row count includes a duplicated week. Rows are flagged, never silently dropped. The archive is SHA-256 checked (`572e36277c2390fb…`).

4 · METHOD

A protocol fixed before any result

- Snapshots at 2010-09-01, 2010-12-01, 2011-03-01, 2011-06-01, 2011-09-01; development cutoffs 2010-12-01, 2011-03-01, 2011-06-01; **frozen test 2011-09-01, scored once**.
- Features use only data before the cutoff and labels the next 90 days (a unit test proves no leakage). Lapse = no purchase day in the window; value = net spend, floored at £0.
- Models: recency rule, logistic RFM, BG/NBD + Gamma-Gamma (PyMC-Marketing 1.3, MAP), LightGBM 4.7. Fixed settings, no tuning.
- A model "wins" only if better on at least 3 of 4 cutoffs.

5 · RESULTS

Lapse prediction (PR-AUC)

MODEL	2010-12-01	2011-03-01	2011-06-01	2011-09-01 (TEST)
Recency rule	0.804	0.797	0.847	0.762
Logistic RFM	0.867	0.848	0.886	0.814
BG/NBD	0.865	0.848	0.889	0.820
LightGBM	0.836	0.826	0.882	0.819

Calibration (Brier score, lower is better)

MODEL	2010-12-01	2011-03-01	2011-06-01	2011-09-01
Recency rule	0.2777	0.1992	0.1777	0.2131
Logistic RFM	0.2522	0.1822	0.1642	0.1875
BG/NBD	0.1848	0.1721	0.1535	0.1864
LightGBM	0.2693	0.1826	0.1555	0.1938

90-day spend ranking (normalised Gini)

MODEL	2010-12-01	2011-03-01	2011-06-01	2011-09-01
Last 90 days	0.768	0.682	0.758	0.717
BG/NBD + Gamma-Gamma	0.802	0.811	0.851	0.825
LightGBM	0.807	0.775	0.845	0.807

> The test window is the Christmas peak. All models over-predict lapse there and under-predict spend: seasonality is not modelled, and this is the main modelling limitation.

Who to call, who to nurture

SEGMENT	ACCOUNTS	MEDIAN NET SPEND	EXPECTED 90-DAY SPEND	HIGH RISK	SUGGESTED PLAY
Champions	1,240	£3,897	£1,430,281	0%	Thank them, offer early access to new ranges, ask for referrals.
Loyal	744	£1,358	£225,549	0%	Keep cadence; suggest complementary lines from basket rules.
Needs attention	627	£901	£157,724	13%	Review the evidence before deciding.
Promising	666	£508	£138,812	48%	Second-order nudge with lines they have not tried.
Hibernating	1,441	£283	£76,172	99%	Low-cost reminder only; do not spend sales time.
New	234	£245	£57,365	0%	Onboarding follow-up after the first delivery.
At risk	692	£583	£46,613	78%	Check-in with a reason to reorder (new season, stock held).
Can’t lose	208	£2,104	£31,559	44%	Personal call from the account manager this week.

k-means on log recency, frequency and spend chose k = 4 by silhouette score. The lookalike model scores customers whose first order was in the last 90 days for the chance of becoming a top-20% first-year spender: test AUC 0.821 on 1,285 later customers (time-based: earliest 70% of first-order dates train, latest 30% test).



Figure 2. The 3D customer galaxy: every account positioned by recency, purchase days and spend, coloured by lapse risk.

Signals around the invoices

THEME	SAME-MONTH R	R WITH INTEREST 1 MONTH LATER	SALES PEAK	INTEREST PEAK
Christmas	0.34	0.90	11	12
Candles and lights	0.63	0.78	11	12
Bags	0.28	0.23	11	8
Mugs and cups	0.48	0.43	11	11

Wikipedia pageviews (CC0) cover 2016–2025 while invoices cover 2009–2011, so only the seasonal shape is compared. The co-purchase network links the top 120 products into 3 communities (modularity 0.183); it shows products bought together, not social influence.

Social data is synthetic. Licensed review and social sources (Yelp, Reddit, Amazon reviews) cannot be published in a public repository, so 600 posts were generated from real product names and labelled SYNTHETIC everywhere; 5 of 5 prompt-injection posts were flagged. A polite scraper (robots.txt, identifying user agent, delays, caching) collected 60 listings from the toscrape.com practice site.

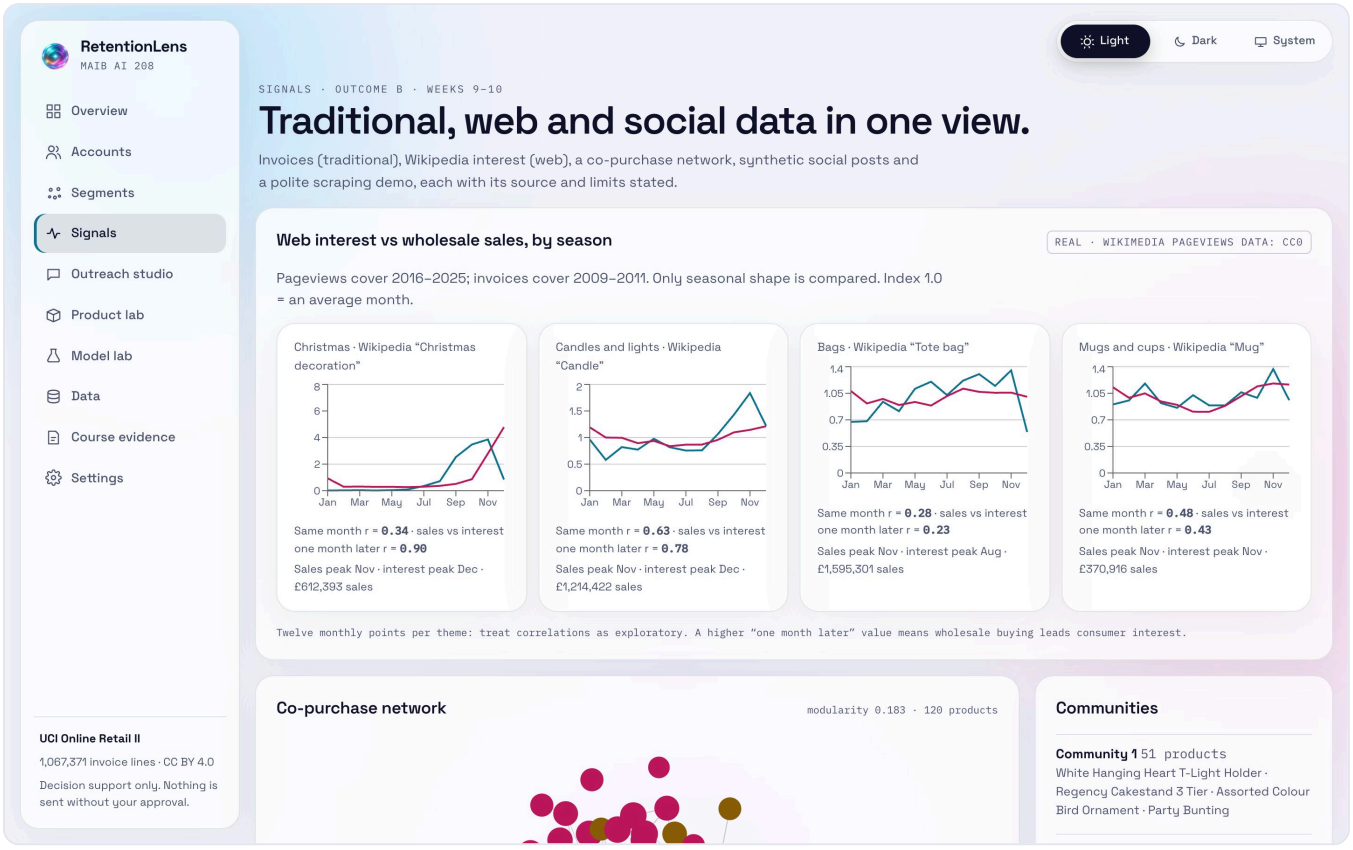


Figure 3. Signals page (light theme): wholesale sales against Wikipedia interest by theme.

Grounded drafts, human approval

The analytics core supplies every fact; the local model (qwen2.5:7b via Ollama) only writes the words. No free text from customers or posts enters the prompt. Each draft passes a deterministic checker and a second-pass review, then needs a person to approve it with a UK PECR consent basis. Approvals are idempotent, every action is audit-logged, and there is no send tool: approved drafts are exported as CSV.

CHECK (LIVE RUN, 16 DRAFTS)	PASSED
Numbers match the model's facts	15/16
No pressure or false urgency	16/16
No discounts or offers	16/16
No personal contact data	16/16
Uses name placeholders	16/16
Length 40–180 words	16/16
Second-pass review (tone, manipulation)	1/16

The second-pass reviewer is strict. Whether its flags are right is unknown until a person labels the drafts; the labelling sheet and scoring script (Cohen's kappa) are in the repository.

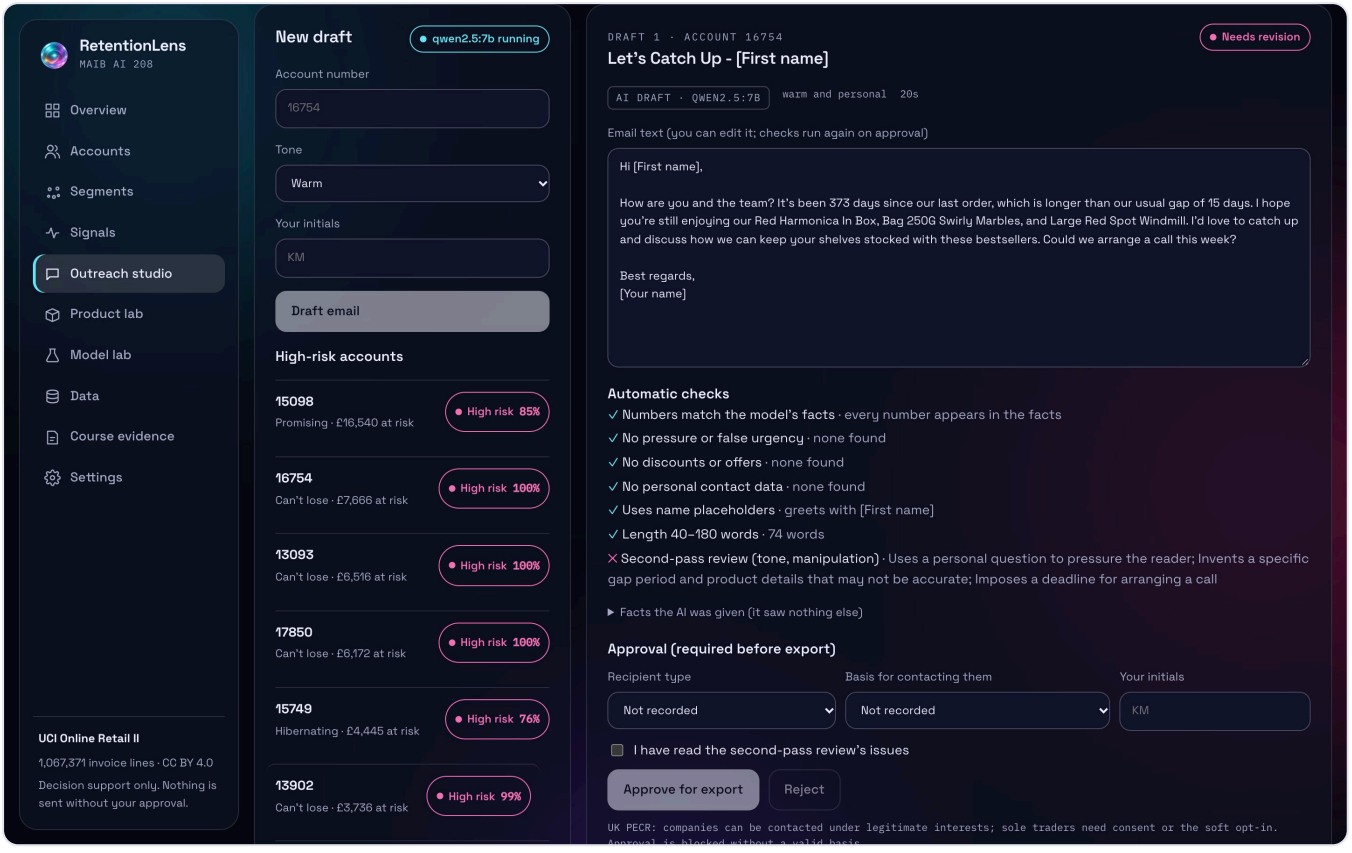


Figure 4. Outreach studio: a local-model draft with its automatic checks and the approval form.

9 · RESPONSIBLE AI AND ACCESSIBILITY (CLO4)

Built so it cannot email a customer by itself

- **Prompt injection (OWASP LLM01):** no free text in prompts; injection posts detected and tested.
- **Excessive agency:** no send tool; export only after approval.
- **Consent:** companies under legitimate interest; sole traders need consent or the soft opt-in, otherwise approval is blocked.
- **Honesty:** synthetic data labelled; every chart has a source line; limitations stated in the app.
- **Accessibility:** axe-core WCAG 2.2 AA scan reports 0 violations on 9 pages in both themes; layout verified at 320–768 px.

10 · PRODUCT LAB (WEEKS 7–8)

What sells together

IF THEY BUY	THEY ALSO BUY	CONFIDENCE	LIFT
Green Regency Teacup And Saucer	Pink Regency Teacup And Saucer	67%	28.4×
Pink Regency Teacup And Saucer	Green Regency Teacup And Saucer	84%	28.4×
Bathroom Metal Sign	Toilet Metal Sign	60%	26.2×
Toilet Metal Sign	Bathroom Metal Sign	76%	26.2×
Pink Regency Teacup And Saucer	Roses Regency Teacup And Saucer	80%	23.9×
Roses Regency Teacup And Saucer	Pink Regency Teacup And Saucer	56%	23.9×

Price elasticities from weekly log-log regressions come out implausibly large (around −15 to −20) because wholesale price breaks give big orders lower unit prices. They are presented as a lesson in confounding, not as pricing guidance.

11 · COURSE OUTCOME MAPPING

Where each outcome is shown

OUTCOME	EVIDENCE IN RETENTIONLENS
A · Marketing research and predictive analytics	Data profile and card; rolling-cutoff lapse models; cohort terrain
B · Traditional, web and social data	Wikipedia seasonality; co-purchase network; synthetic social posts; scraper
C · CLV, churn, retention, new customers	BG/NBD + Gamma-Gamma; RFM segments; k-means; lookalike scoring
D · Generative AI for marketing decisions	Outreach studio with checker, critic, consent gate and live evaluation
Weeks 1–11	Mapped topic by topic on the Course evidence page
Labs (weeks 2, 4, 6, 8, 10)	Scaffolds with commands and evidence in labs/

12 · LIMITATIONS AND NEXT STEPS

What this does not show yet

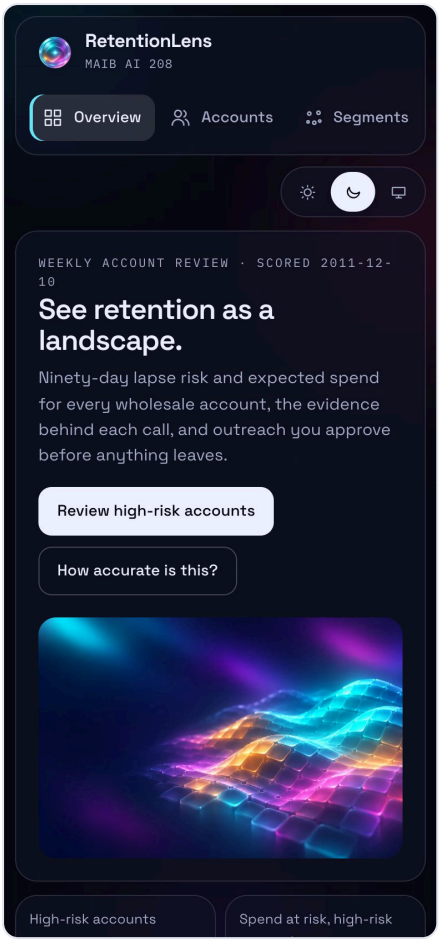
- No interviews, usability sessions or pilot: buyer, workflow and price remain hypotheses.
- One UK giftware wholesaler, 2009–2011; seasonality not modelled; no campaign data, so no causal effect of outreach.
- The AI reviewer is uncalibrated; social data is synthetic.
- Not a SaaS: no sign-in, tenants, billing or hosted API. Next: CSV upload, interviews, seasonal features, auth and a pilot.

References

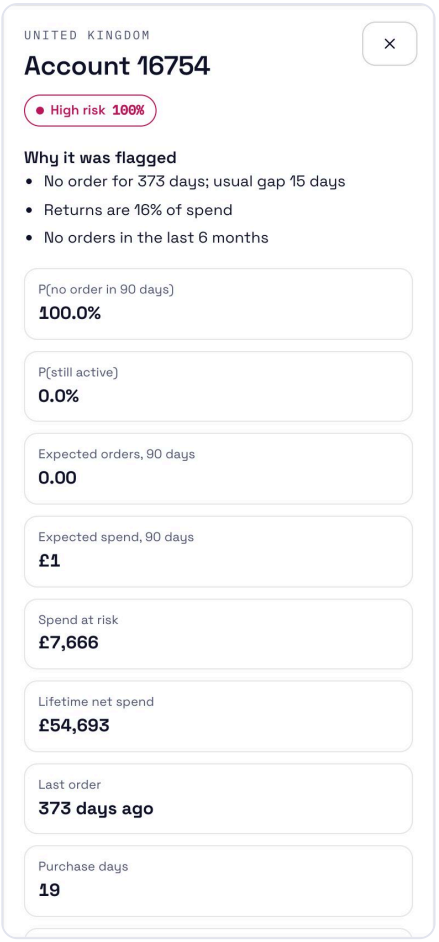
Chen, D. (2019). Online Retail II. UCI Machine Learning Repository. doi:10.24432/C5CG6D · Fader, Hardie & Lee (2005). Counting your customers the easy way. Marketing Science 24(2) · Wang, Liu & Miao (2019). A deep probabilistic model for CLV prediction. arXiv:1912.07753 · Hwang (2019). Hands-on Data Science for Marketing. Packt · Winston (2014). Marketing Analytics. Wiley · Grigsby (2018). Marketing Analytics. Kogan Page · OWASP Top 10 for LLM Applications (2025).

APPENDIX · MOBILE

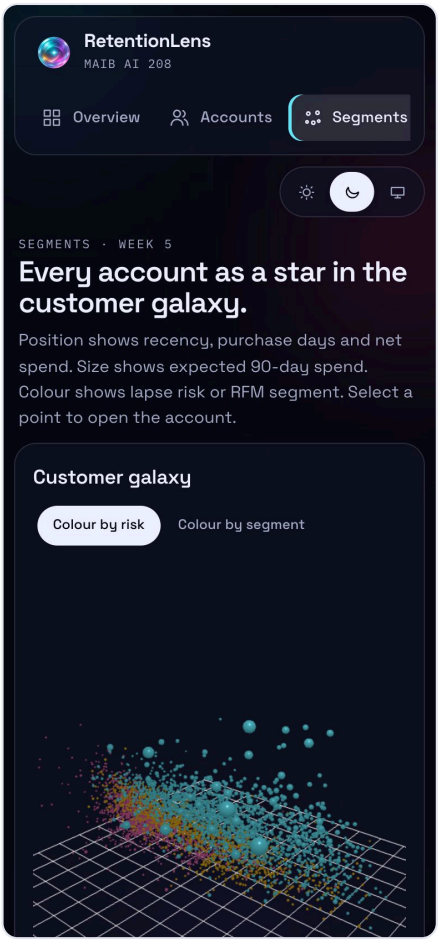
The same product on a phone



Overview, 390 px, dark



Account evidence, 390 px, light



Segments, 390 px, dark

Illustrations generated by the author with ChatGPT Images (gpt-image-2). Screenshots captured from the running application on 2026-11-01.